



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E EEE 'A'

Course Code & Title: EE8391 – Electromagnetic Theory

Name of the Faculty member: Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit I / Applications of the Gauss's law
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO 4
- **Activity Chosen:** Think-Pair-Share
- **Justification:**
 - The topic "Application of Gauss law" demands that the students have to analysis and apply standard defined procedure to compute electric field intensity based on their understanding of the Gauss law related concepts. Hence it is like finding the solution for a problem, I have chosen the collaborative learning technique "Think-Pair-Share (TPS)".
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
 - The students have been grouped into smaller groups of four members, with one bright student, two average learners and one slow learner in each group. One average learner from each group has been made as a recorder.
 - I have shown the power point presentation for the quick recap of the concepts like Electric field intensity, Charge distributions, Gaussian surface, properties of Gaussian surface and Gaussian theorem.
 - The detailed step by step procedure has been explained to the students.
 - The students have been told to think and work out the solution individually for 1 minute and then asked to discuss about their solution within the group for 1 minute.
 - Finally, two recorders (one for each problem) have been asked to share their results to the whole class through chalk and talk technique. All other students interacted with them for correctness of their answers and different approaches for the same problem.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 12	PSO 1
CO 1	3	2	1	1	1

(1 – Low

2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10
	1	1
Justification for correlation	The students will do this activity in team. Hence it is mapped with PO 9 with level 1.	The students will communicate with each other and do the oral presentation. Hence this activity is mapped with PO 10 with level 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the activity. Most of the students will fear when they have to apply mathematical equations to electrical engineering concepts. But when I did this activity, they never worried about learning the mathematics part and had a great confidence in learning the new concept.

- ❖ ***Benefit of the practice:***

- ✓ The students have understood the concepts clearly and it has been evident from the points they have shared with the classroom.
- ✓ The students have presented the points to the class; it cleared any ambiguity in understanding the important points in the chosen topic.

- ❖ ***Challenges faced in implementation:***

- ✓ The activity has been planned for 15 minutes, but it has been completed after 20 minutes only. Next time, I should plan the activity with proper time.
- ✓ The group has to form as heterogeneous, and then only I can take care about the non-participating students.

References:

- ❖ Mathew N. O. Sadiku, 'Principles of Electromagnetics', 6th Edition, Oxford University Press Inc. Asian edition, 2015.
- ❖ William H. Hayt and John A. Buck, 'Engineering Electromagnetics', McGraw Hill Special Indian edition, 2014.
- ❖ <https://www.readingrockets.org/strategies/think-pair-share>

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Course Code & Title: EE8391 – Electromagnetic Theory

Name of the Faculty member: Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit I & II / Point charge, Electric field, Electric flux, Electric field intensity calculations, Electric potential & Equipotential surfaces
- **Course Outcome:** CO 1 & CO 2
- **Topic Learning Outcome:** TLO 3, TLO 4, TLO 5
- **Activity Chosen:** Virtual Lab
- **Justification:**
 - The PhET tool has a simulation for the electric field which can be used for explaining the concepts of Point charge, Electric field, Electric flux, Electric field intensity calculations, Electric potential & Equipotential surfaces. Hence this Virtual laboratory has been created and used for the explaining these topics.
- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**
 - The students are taken to the Power system simulation laboratory and each one is occupying the one system with Internet connectivity.
 - The link for the simulation is shown to them through the LCD projector and I told them go to that page.
 - I have provided the soft copy of the lab manual prepared and explained about the virtual experiments and how to do those experiments step by step. I also told my students to do the same parallel.
 - Finally, the students have taken the screenshots of their laboratory experiment setup and resultant waveform and submitted as an Assignment in CANVAS.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 8	PO 12	PSO 1
CO 1	3	2	1	-	1	1
CO 2	3	2	1	1	1	1

(1 – Low

2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 10
	1
Justification for correlation	The students are writing the report for the Virtual laboratory experiments. Hence their written communication skills are getting improved. Hence it is mapped with level 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the hands-on experience on electric field and potential visualization.
- ✓ These concepts are learned only through theory-based lectures. But when I proposed this activity as an Assignment, they really liked it and did it very well.

❖ ***Benefit of the practice:***

- ✓ The students are able to visualize about the electric fields virtually. It is made them to understand the concepts very well.
- ✓ Since it is virtual environment, the students are able to do the experiment multiple times until they understand the topic.

❖ ***Challenges faced in implementation:***

- ✓ Few students had a problem to navigate through the PhET tool. I helped them to do it.
- ✓ Managing the students, making them to do the experiments and clearing the doubts about the tool and concepts took lot of time. Hence, it is not possible to cover all the experiments in one period.

References:

- ❖ Mathew N. O. Sadiku, 'Principles of Electromagnetics', 6th Edition, Oxford University Press Inc. Asian edition, 2015.
- ❖ William H. Hayt and John A. Buck, 'Engineering Electromagnetics', McGraw Hill Special Indian edition, 2014.
- ❖ https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html

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Course Code & Title: EE8391 – Electromagnetic Theory

Name of the Faculty member: Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit III / Classification and types of magnetic materials
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Visible Quiz
- **Justification:**
 - The chosen topic will have classes of the magnetic materials and each material will have its own characteristics. Usually the students will find it difficult to be attentive during this topic. By conducting this visible quiz, I tried to help them to understand the concepts better and also the students will be more interested in doing Quiz.
- **Time Allotted for the Activity:** 15 minutes
- **Details of the Implementation:**
 - The students are taken to the Power system simulation laboratory and each one is occupying the one system with Internet connectivity.
 - The quiz has been created in the Quizizz online tool and pass code has been shared with the students.
 - All the students have started the quiz at the same time. At the end of the quiz, the correct answers will be shown to them in the screen.
 - Finally, I explained the correct answer for the quiz questions with proper reasons.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 9	PO 10	PO 12	PSO 1
CO 3	3	2	1	1	1	1	1

(1 – Low

2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 5	PO 8
	1	1
Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 1	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the quiz. They told they have been given with the chance of remembering the concepts in the same class itself.

- ❖ ***Benefit of the practice:***

- ✓ The students are getting the feel of involved in the learning process.
- ✓ Since the quiz is not included in the marks calculation, the students have done the quiz without any test fear. Hence, they participated freely.

- ❖ ***Challenges faced in implementation:***

- ✓ Few students have not much interest in attending the quiz sincerely, since the quiz is ungraded. Managing those students and making them to do the quiz is difficult task for me in this activity.

References:

- ❖ Mathew N. O. Sadiku, 'Principles of Electromagnetics', 6th Edition, Oxford University Press Inc. Asian edition, 2015.
- ❖ William H. Hayt and John A. Buck, 'Engineering Electromagnetics', McGraw Hill Special Indian edition, 2014.
- ❖ https://prezi.com/9_lavlte_bbr/visible-quiz/

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Name of the Faculty member: Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic:** Unit IV / Maxwell's Equations
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 15
- **Activity Chosen:** One minute paper
- **Justification:**
 - The chosen topic is of derivative in nature with lot of mathematical and engineering concepts involved. This activity helps the students to reflect on the day's lesson in response to my question and helps them in remembering and recollecting the important points.
 - The activity provides me with useful feedback about how the students have understood my concepts.
 - One minute paper is an in-class writing activity taking one-minute or less to complete. Hence it is easily implementable.
- **Time Allotted for the Activity:** 5 minutes
- **Details of the Implementation:**
 - At the end of the lecture (about 40 minutes), the students are asked to write down the Maxwell's equations along with the corresponding laws that they learned during the class in a piece of paper. The paper has been given to the students by the teacher.
 - The students have been given with three minutes of time. They need to write the points learned in the class in the paper.
 - At the end of the 3 minutes, all the papers are collected by the teacher. Randomly few papers are selected. The points written on it are discussed to the class by the students themselves. Teacher facilitated the process and added the explanation to the student points.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 5	PO 8	PO 12	PSO 1
CO 4	3	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 10
	1
Justification for correlation	The students are writing the points and explaining the points to the class. Hence their communication skills are getting improved. Hence it is mapped with level 1

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ✓ The students have enjoyed the session even though they have been asked to write the important points on the paper. I told them to consider it as a practice not as a test. Hence they felt calm and did the work.

- ❖ ***Benefit of the practice:***

- ✓ The students have understood the concepts clearly and it has been evident from the points they have written on the one minute paper.
- ✓ The students have presented the points to the class; it cleared any ambiguity in understanding the important points in the chosen topic.

- ❖ ***Challenges faced in implementation:***

- ✓ Managing the non-participating students in the class. Mostly the slow learners have not participated in the activity. They have not written any points on the paper.
- ✓ When I discussed with them, they said they had a fear of asking points to them. Hence they have not participated.

References:

- ❖ Mathew N. O. Sadiku, 'Principles of Electromagnetics', 6th Edition, Oxford University Press Inc. Asian edition, 2015.
- ❖ William H. Hayt and John A. Buck, 'Engineering Electromagnetics', McGraw Hill Special Indian edition, 2014.
- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>

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Name of the Faculty member: Dr. K. Karthikeyan

Innovative Practice Description

- **Unit / Topic: Unit V / Electromagnetic wave propagation in various media**
- **Course Outcome: CO 5**
- **Topic Learning Outcome: TLO 18**
- **Activity Chosen: Students Teams Achievement Division (STAD)**
- **Justification:**
 - The selected topic comprises of lot of derivations and a theory concept which is generally students feel very difficult to understand.
 - The STAD is most suitable for the topics having a mixture of theory concepts and mathematical derivations.
 - The electromagnetic wave will exhibit various phenomena while travelling in various media. Hence the students have to understand and compare the behavior of the wave in various media. This will make the students to feel it is difficult for them to learn.
 - The collaborative learning or peer coaching can be easily done by the students for the theory concepts along with derivations rather than for the problematic subjects. Hence STAD has been chosen for this topic.
- **Time Allotted for the Activity: 2 Periods**
- **Details of the Implementation:**
 - The Initial quiz to pre check their knowledge on the topic to be learned has been conducted with 10 questions with simple and medium level of complexity.
 - Students are allowed to prepare with group for the quiz questions uploaded in the course website.
 - Each group has to explain about one media. Randomly I will ask the students few questions for better understanding of the concepts. During this process, other members can help to give the answer.
 - Again Online quiz for individual students with Quiz will comprising of 30 questions of various levels which will cover the entire lesson.
 - Finally the scores and correct answers are displayed to the students and the explanation of the correct answers is given to the students.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 12	PSO 1	PSO 3
CO 1	3	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9	PO 10
	1	1
Justification for correlation	The students are working in a group while doing this activity. Hence it is mapped with level 1.	The students are writing the answers and explaining the answers to the class. Hence their communication skills are getting improved. Hence it is mapped with level 1.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ The students are comfortable in this activity and they fully enjoyed the activity. They told their understanding level of the topic has improved and it gave them an opportunity to communicate with their team members.

- ❖ **Benefit of the practice:**

- ✓ The students have understood the concepts clearly and it has been evident from the answers they have given for the Quiz.
- ✓ The slow learners have utilized the fast learners help and understood the concepts very well.

- ❖ **Challenges faced in implementation:**

- ✓ The planning has to be done in a more effective way. The activity has to be planned with different active learning techniques. Other suitable active learning technique has to be explored and used for planning the next activity.
- ✓ The time allotment has to be thoroughly changed. Each and every activity needs more time to complete.
- ✓ The Assessment methods are needed to be modified, the different formative and summative assessment procedure to be studied and explored.

References:

- ❖ Mathew N. O. Sadiku, 'Principles of Electromagnetics', 6th Edition, Oxford University Press Inc. Asian edition, 2015.

- ❖ William H. Hayt and John A. Buck, 'Engineering Electromagnetics', McGraw Hill Special Indian edition, 2014.
- ❖ Tiantong, Monchai, and Sanit Teemuangsai. "Student Team Achievement Divisions (STAD) Technique through the Moodle to Enhance Learning Achievement." International Education Studies, vol. 6, no. 4, Mar. 2013, p. p85. DOI.org (Crossref), <https://doi.org/10.5539/ies.v6n4p85>.

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